

Phase 2

MERN Stack PROJECT:

Zerodha Trading APP

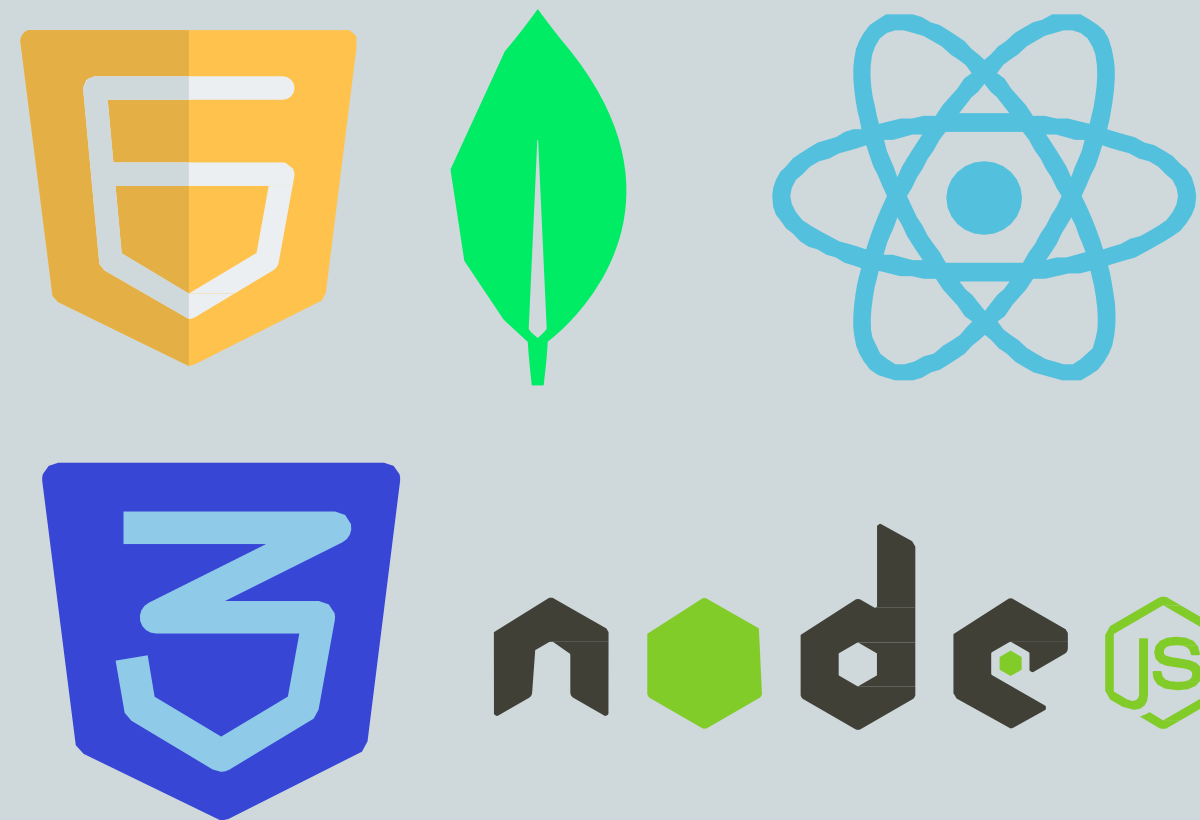
Development Phase

By Zubair Khan

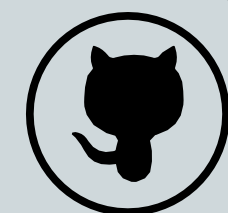
Matriculation: 92127983

BSC in Computer Science

Github Repository: https://github.com/zbr-khn/Zerodha_IU_WebApp



Java & Web Application Development

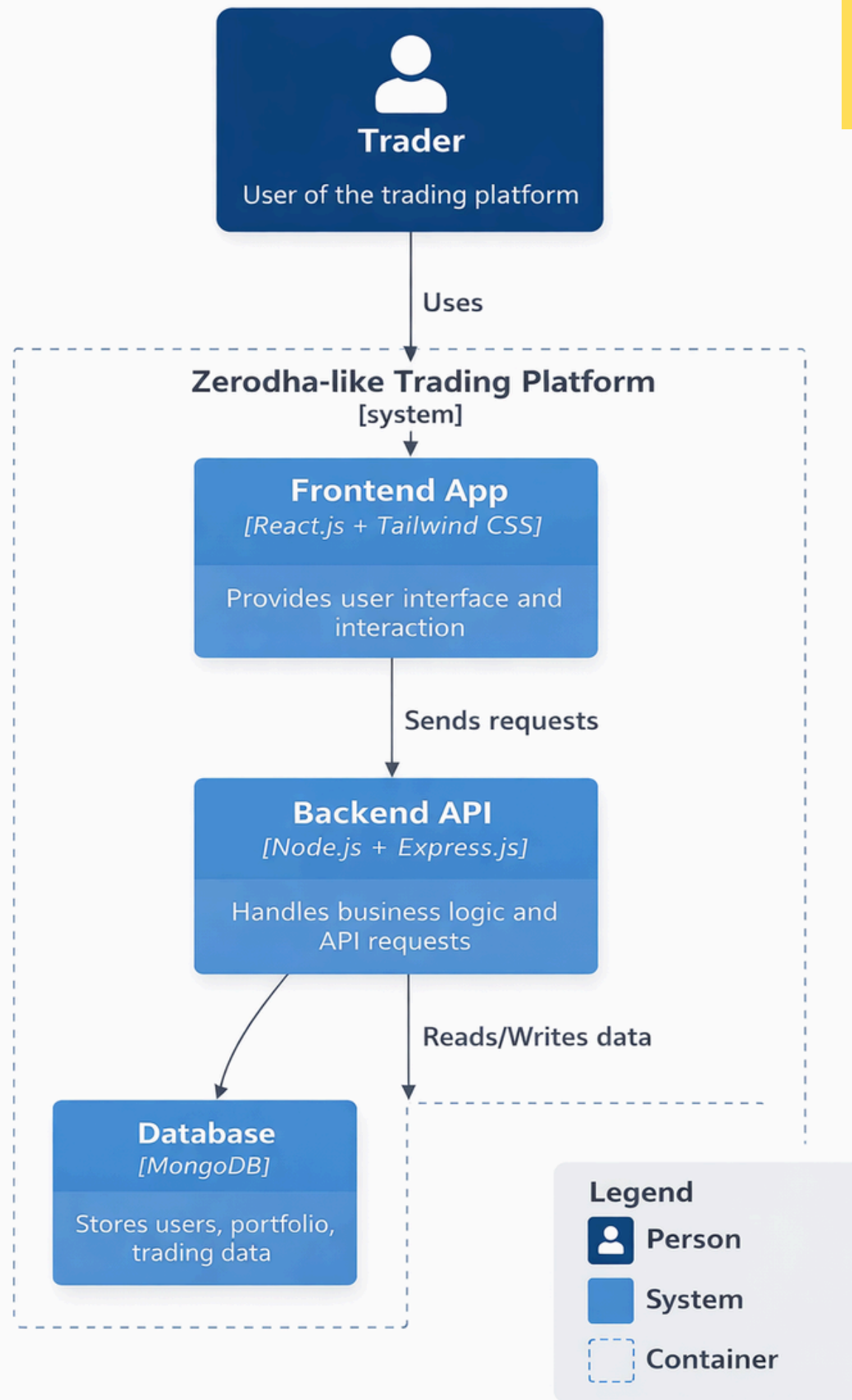


Purpose of the App

The app is designed to give a responsive trading experience in real time, using the **MERN stack** to simulate the environment of a professional online **trader**. It is ideal for anyone who wants to analyze the market, follow their **investments** and use a smooth and secure web-based **trading platform**.

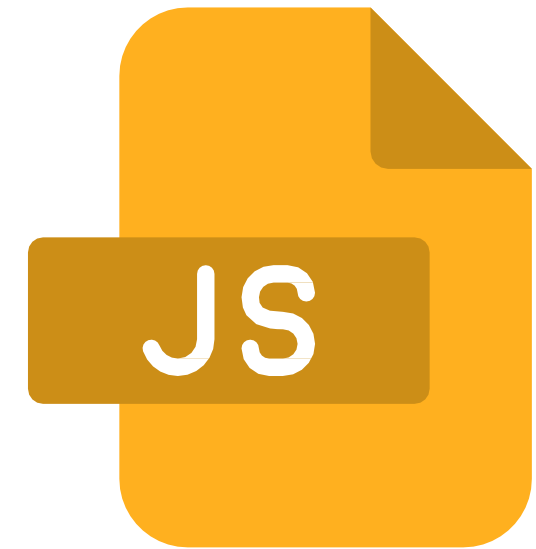
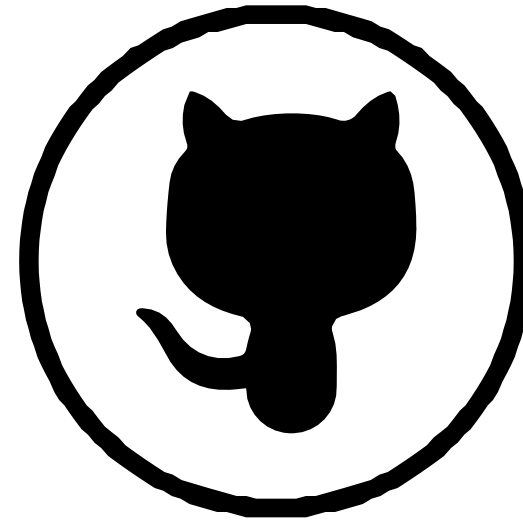
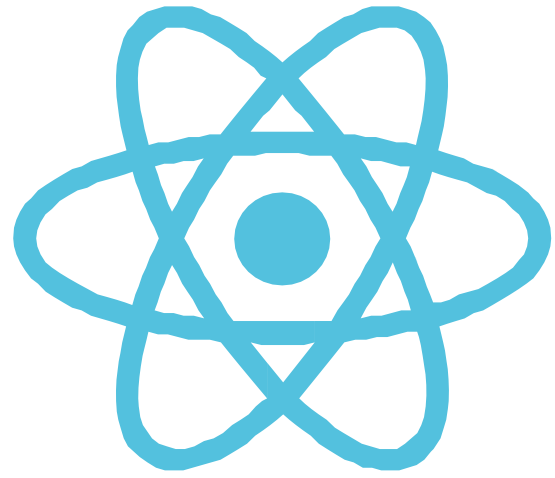
Architecture Diagram

C4 Model





Technology Used



Frontend

React.js (v19.1.0) – Core library for building dynamic UIs
React-Router-Dom (v7.5.3) – Handles client-side navigation
Material UI & Emotion – For styled components and UI icons
Chart.js & React-Chartjs-2 – Enables interactive visual data representation
Axios (v1.6.7) – For API communication with backend services
Web-Vitals – Measures and tracks app performance
Testing Libraries – Integrated with Jest-DOM, User-Event, and React Testing Library for robust frontend testing

Backend (API & Authentication)

Node.js & Express.js (v5.1.0) – Build and serve RESTful APIs
MongoDB & Mongoose (v8.14.1) – For schema-based NoSQL database operations
Passport.js & Passport-Local-Mongoose – Implements secure user authentication
Dotenv – Manages sensitive config via environment variables
CORS & Body-Parser – Middleware for secure and structured request handling
Nodemon – For live development and reloading

Dashboard:

Enhanced with MUI, Chart.js, and React Chart components
Custom insights with visual analytics for portfolios and trading metrics

Changes from the Original Proposal and Reasons

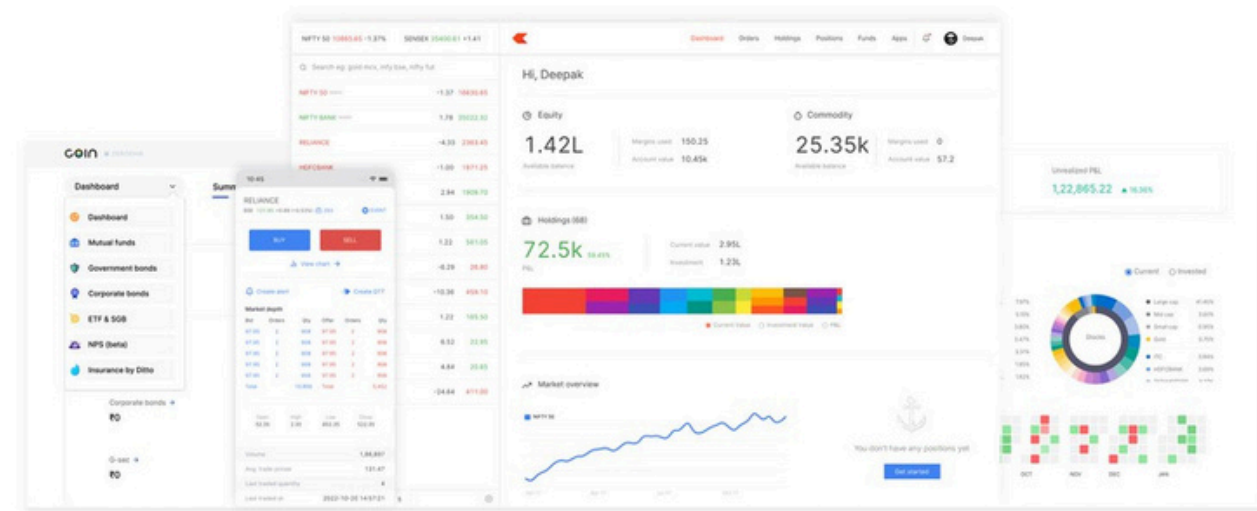
Change	Updated Technology	Justification
React Version Upgrade	React ^19.1.0 (frontend)	Leveraged latest stable release for performance and feature improvements
Routing Enhancement	react-router-dom ^7.5.3 (frontend)	Adopted modern routing API for better navigation handling
UI Framework Integration	MUI + Emotion (dashboard)	Enabled consistent, responsive UI with minimal custom styling
Charting Support Added	chart.js, react-chartjs-2 (dashboard)	Facilitated visual analytics for financial data
HTTP Client Addition	axios (dashboard)	Streamlined API communication and error handling
Backend Stack Implementation	Express, Mongoose, Passport (backend)	Established a robust, scalable, and secure backend API
Auth Integration	passport-local, passport-local-mongoose	Simplified secure user authentication and session management
Config Management	dotenv (backend)	Ensured environment-specific configuration handling
Modern Testing Setup	Updated Testing Library packages	Improved testing reliability and compatibility with latest React
Dev Workflow Enhancement	nodemon (backend)	Enabled auto-reload for efficient backend development

Benefits & Features

- Smooth and user-friendly trading experience that fits any level of investors. Real time market information to trade efficiently and intelligently.
- Deep technical analysis advanced charting.
- Easy and fast order entry to execute trade efficiently. Holding and P&L portfolio monitoring with a detailed view.
- The availability of detailed reports on the history of performance and trading.
- It has fund management functionality that allows easy deposits and withdrawals. Price movement and trade alerts and notifications.
- Kite and Console are examples of integrated tools for increased functionality and analytics.
- One-stop shop to help you invest intelligently and wisely.
-

Home Page

The homepage of Zerodha features the secure, user-first investment platform, which provides a wide variety of financial products, educational resources such as Varsity, and transparent services, which motivate people to invest and learn with ease.



Trust with Confidence

Customer-first always

That's why 1.3+ crore customers trust Zerodha with ₹3.5+ lakh crores worth of equity investments.

No spam or gimmicks

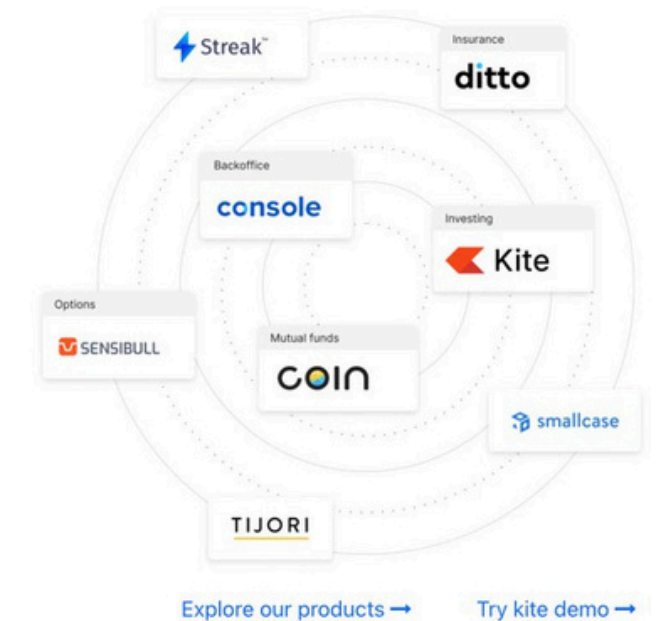
No gimmicks, spam, "gamification", or annoying push notifications. High quality apps that you use at your pace, the way you like.

The Zerodha universe

Not just an app, but a whole ecosystem. Our investments in 30+ fintech startups offer you tailored services specific to your needs.

Do better with money

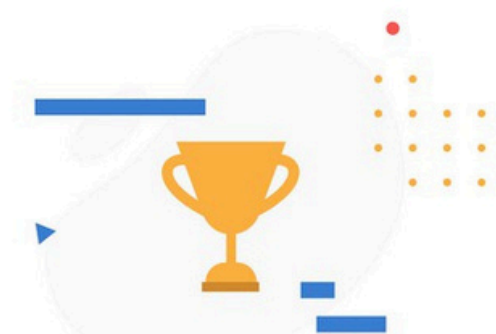
With initiatives like Nudge and Kill Switch, we don't just facilitate transactions, but actively help you do better with your money.



Invest in everything

Online platform to invest in stocks, derivatives, mutual funds and more

[Signup now](#)



Largest Stock Broker in India

2+ million Zerodha clients contribute to over 15% of all retail order volumes in India daily by trading and investing in:

- Futures and Options
- Commodity derivatives
- Currency derivatives
- Stocks & IPOs
- Direct Mutual Funds
- Bonds & Govt. Securities

THE ECONOMIC TIMES Forbes Business Line THE HINDU mint

Economic Times Startup of the Year - 2020
NSE, BSE, MCX - Best Retail Brokerage award - 2018, 2019 & 2020

Free and open market education

Varsity, the largest online stock market education book in the world covering everything from the basics to advanced trading.

[Varsity](#)

TradingQ&A, the most active trading and investment community in India for all your market related queries.

[TradingQ&A](#)

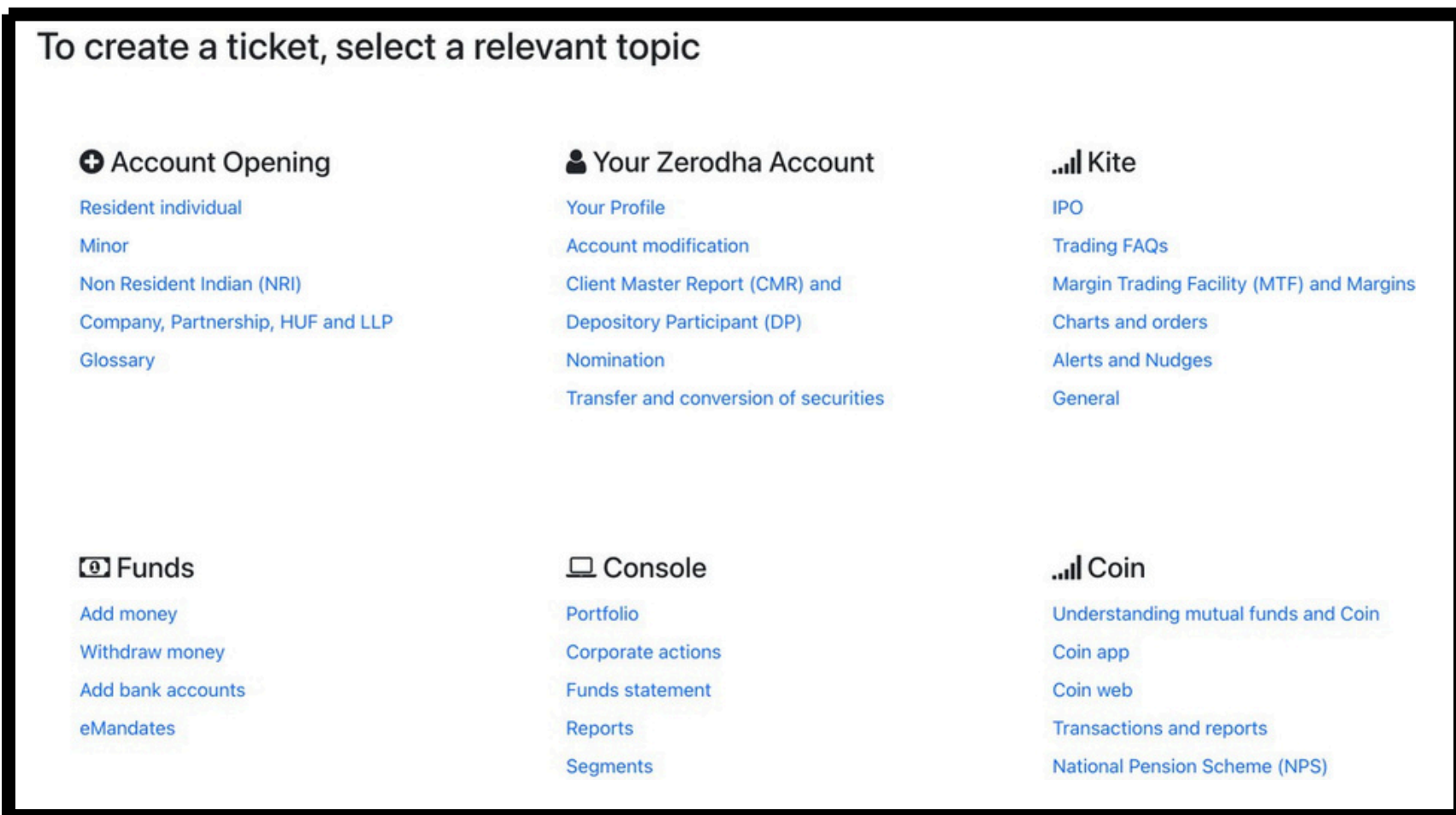
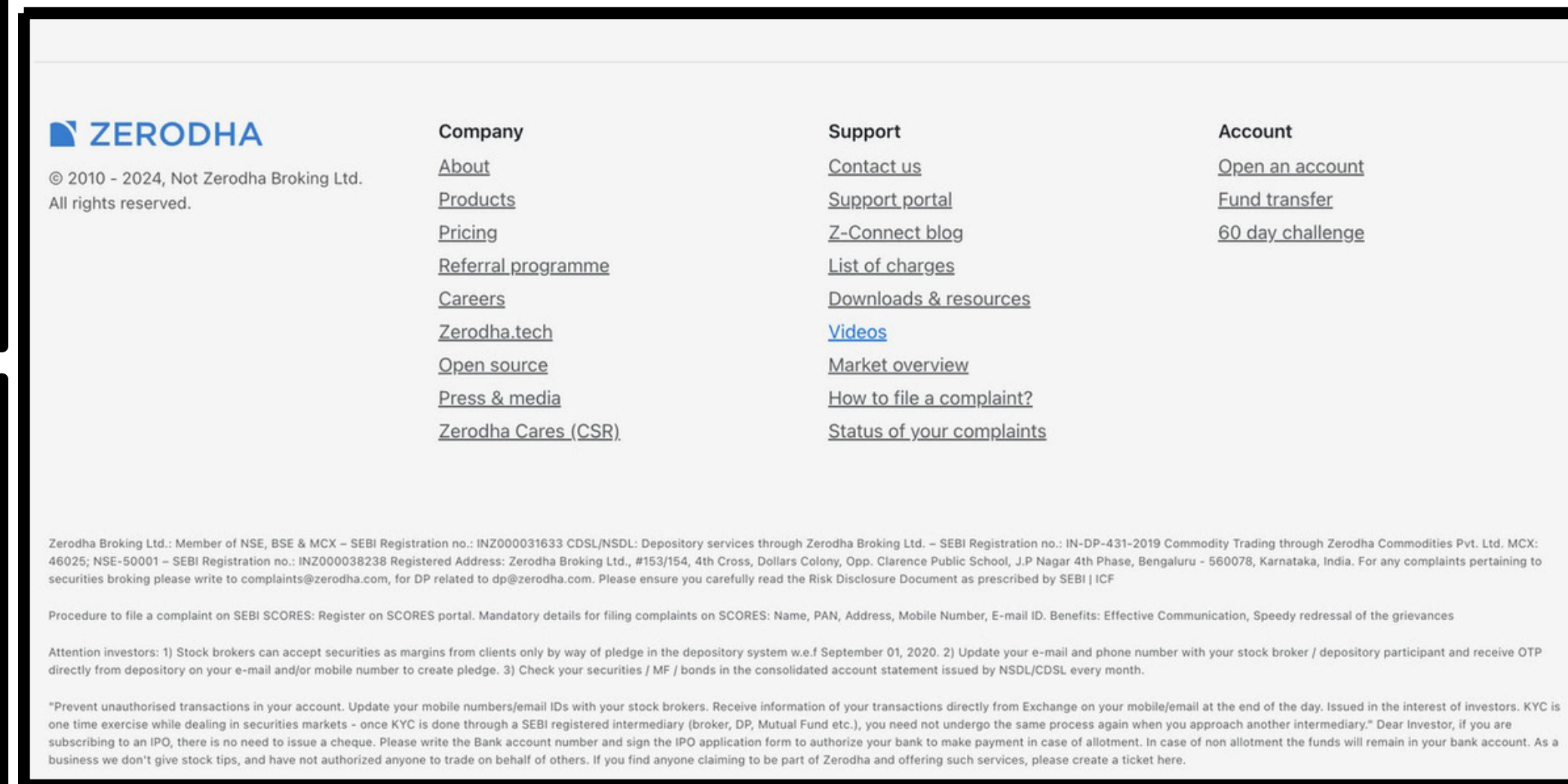
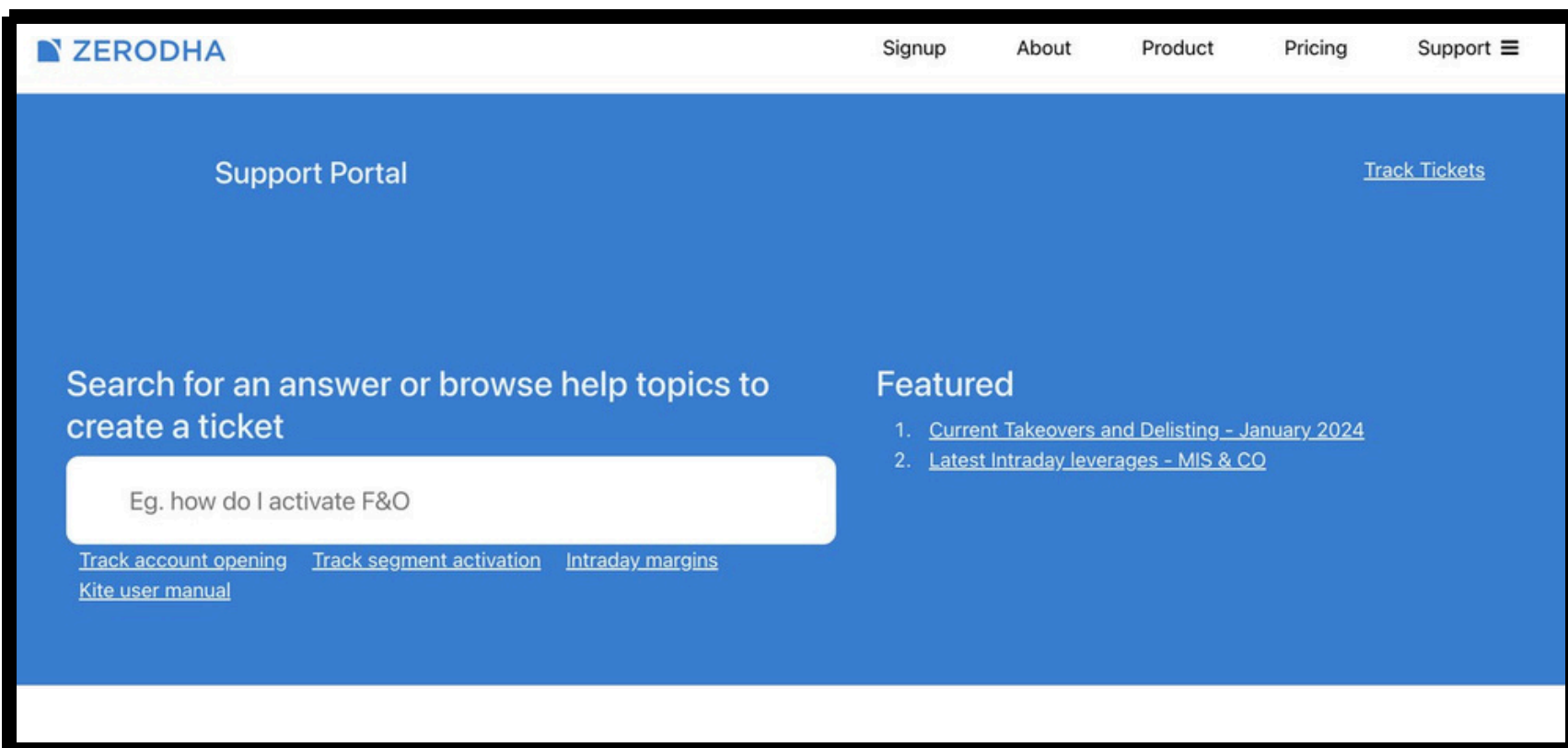
Open a Zerodha account

Modern platforms and apps, ₹0 investments, and flat ₹20 intraday and F&O trades.

[Sign up now](#)

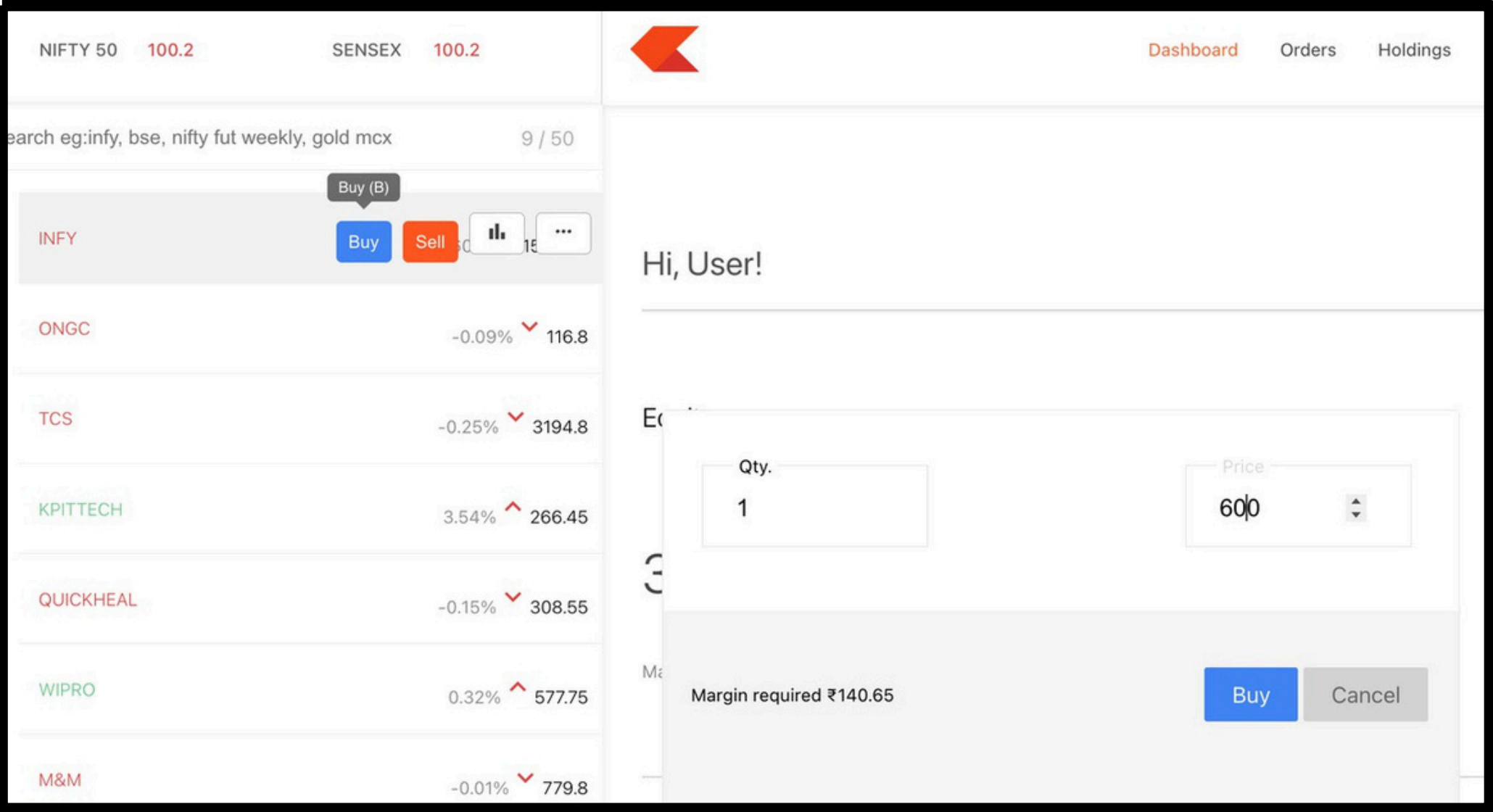
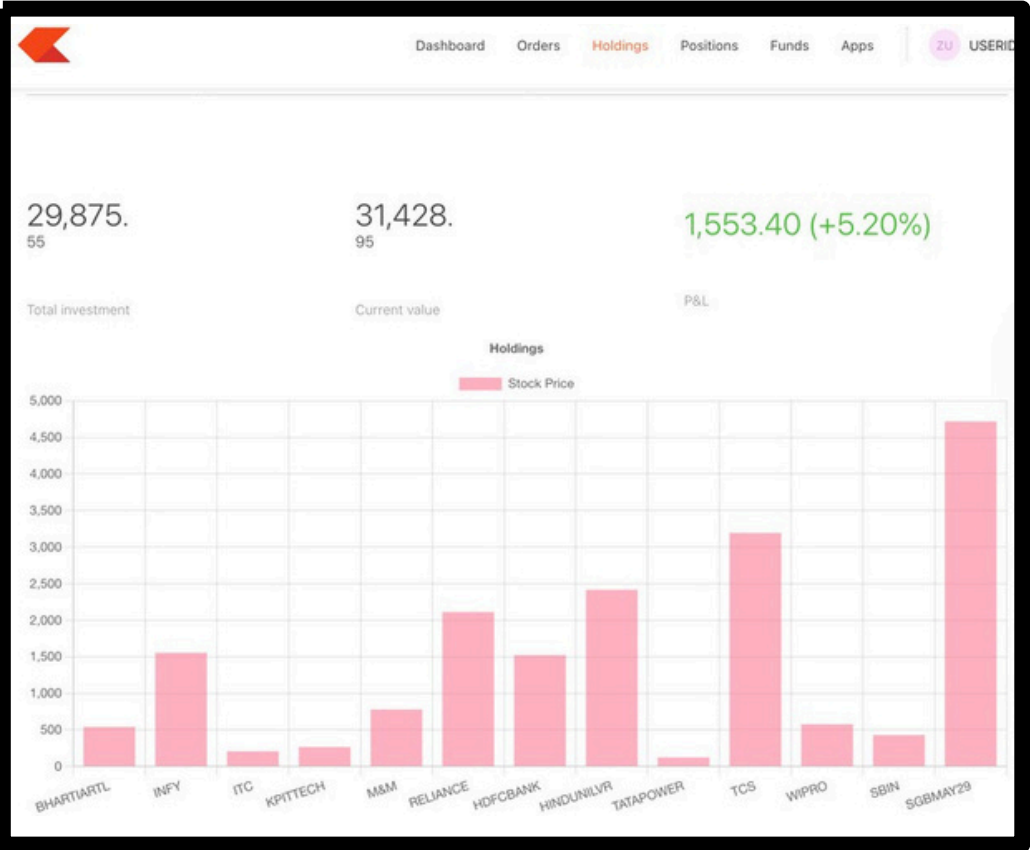
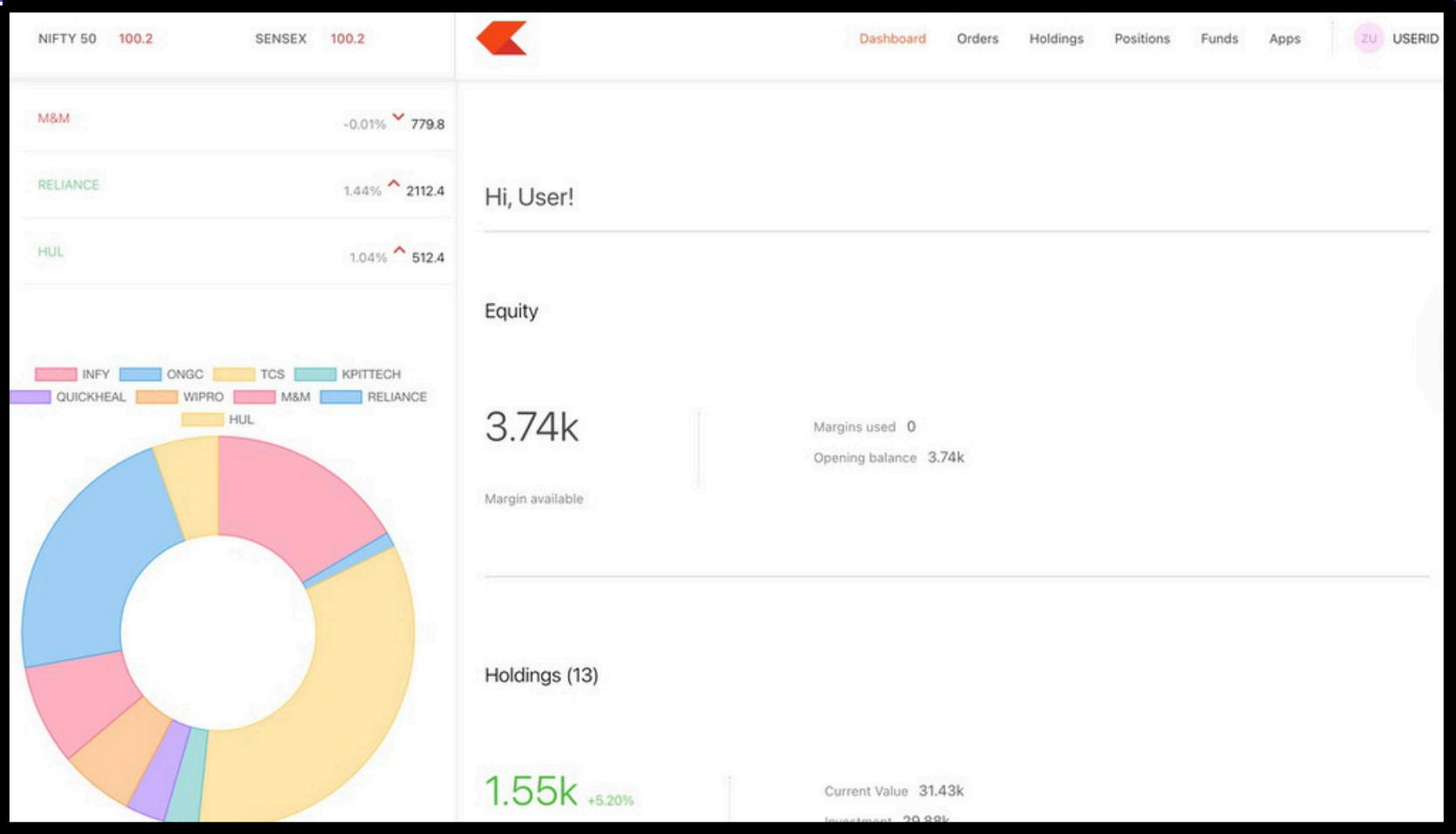
Support Portal

Zerodha support portal provides a focal point of user help-it provides help topics, creation of tickets, and support categorized to speed up an issue.



Dashboard

The dashboard provided by Zerodha gives a real-time and complete view of the trading portfolio of the user. The interface shows major data including equity, holdings and profit/loss overviews, as well as easy to understand visual charts, such as pie and bar graphs. The dashboard will allow users to monitor their stocks and see market indices and also make trades. It facilitates fast accessibility to orders, positions and funds, which guarantees a smooth portfolio tracking and trading decisions.



Test Case 1

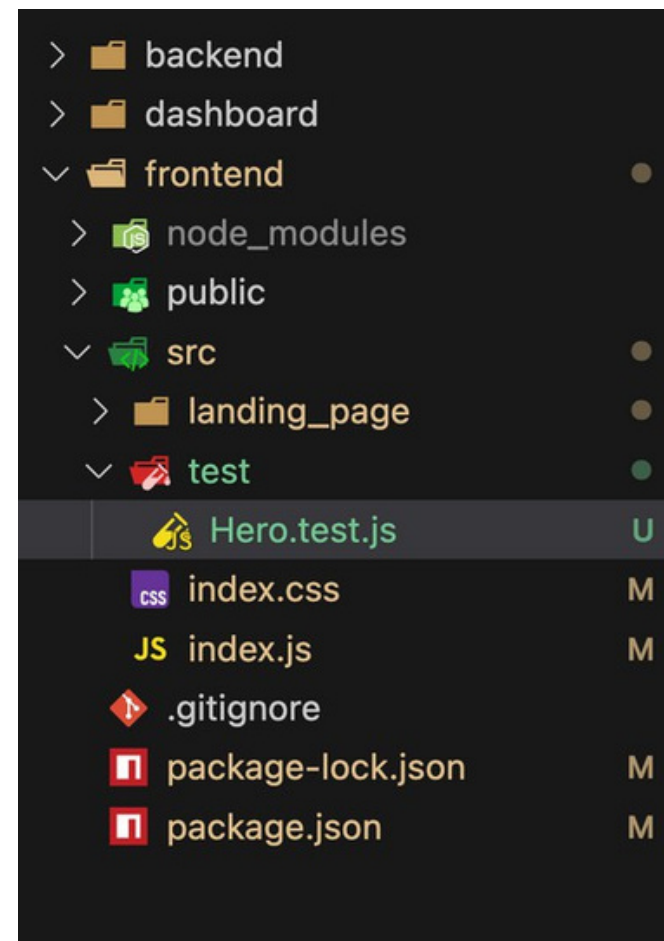
In case of the Zerodha Trading App, the component testing was applied with the help of Jest and React Testing Library to make sure that the major UI elements appear and work as intended. Emphasis was put on the Hero part of the landing page.

Two large tests were developed:

Image Rendering Test: This tests that the Hero image is loaded properly with use of alt text, and that the image source attribute indicates the expected file path.

Signup Button Test: This guarantees that the signup button is visible and serves its purpose. It also confirms that the appropriate styling class (btn-primary) is present, guaranteeing a consistent look and ease of use.

The tests make the UI more reliable and contribute to the early detection of regressions during development.



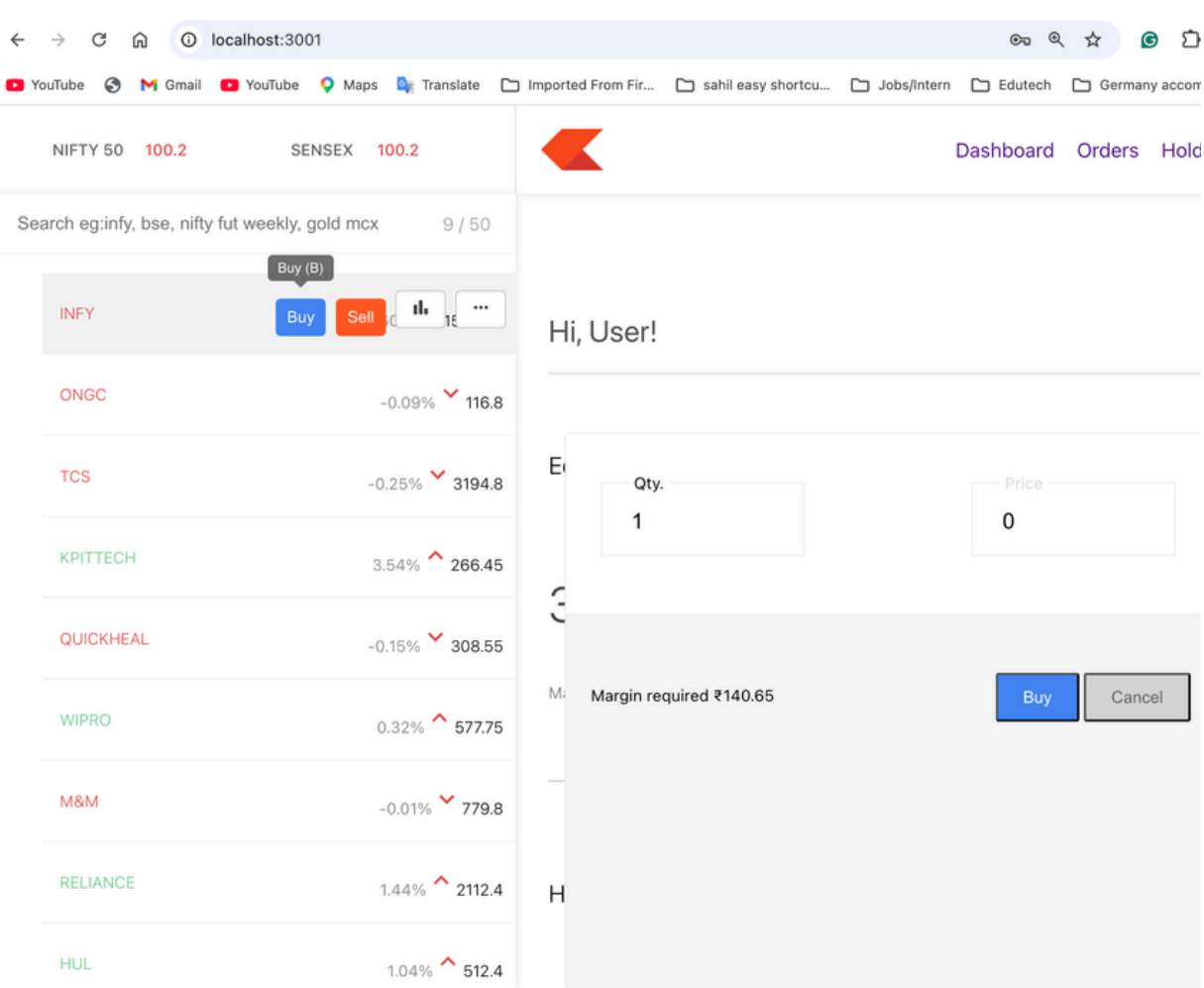
```
Hero.test.js U X
frontend > src > test > Hero.test.js > describe('Hero Component') callback
1  import React from 'react';
2  import {render, screen} from '@testing-library/react';
3  import '@testing-library/jest-dom';
4
5  import Hero from '../landing_page/home/Hero';
6
7  // Test Suite
8
9  describe('Hero Component', ()=>{
10
11      test('renders hero image', ()=>{
12          render(<Hero/>);
13          const heroImage = screen.getByAltText("Hero Image");
14          expect(heroImage).toBeInTheDocument();
15          expect(heroImage).toHaveAttribute('src', 'media/images/homeHero.png')
16      });
17
18
19      test('renders signup button', () => {
20          render(<Hero />);
21          const signupButton = screen.getByRole('button', { name: /signup now/
22          expect(signupButton).toBeInTheDocument();
23          expect(signupButton).toHaveClass('btn-primary');
24      });
25  })
```

Test Case 2

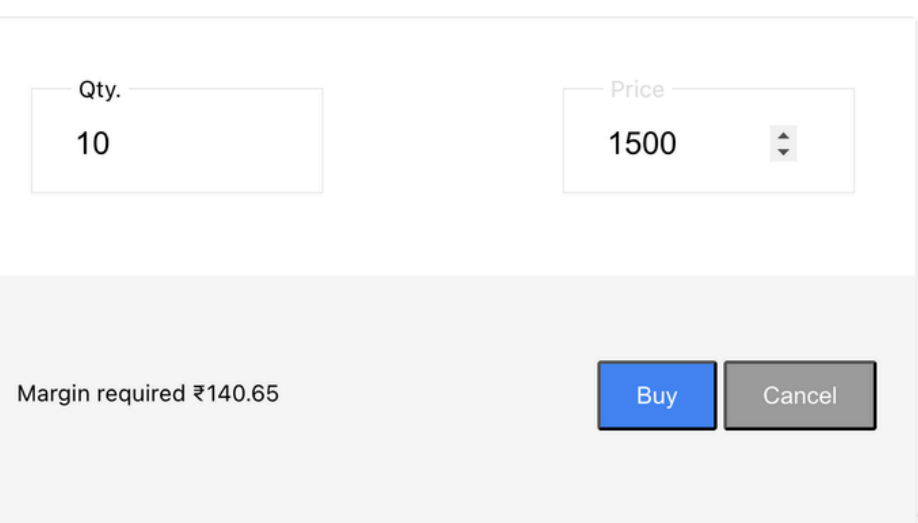
When I buy any new stock, for example

Quantity 10, Price: 1400 then it is coming in the orders section.

Step 1:



Step 2:



Step 3:

				Dashboard	Orders	Holdings	Positions	Funds	Apps	User
ORD-1000	INFY	1	₹300.00	BUY	Executed	---	---			
ORD-1001	INFY	1	₹141.00	BUY	Executed	---	---			
ORD-1002	INFY	1	₹141.00	BUY	Executed	---	---			
ORD-1003	INFY	1	₹141.00	BUY	Executed	---	---			
ORD-1004	INFY	1	₹142.00	BUY	Executed	---	---			
ORD-1005	INFY	1	₹141.00	BUY	Executed	---	---			
ORD-1006	INFY	1	₹141.00	BUY	Executed	---	---			
ORD-1007	INFY	2	₹300.00	BUY	Executed	---	---			
ORD-1008	INFY	1	₹140.65	BUY	Executed	---	---			
ORD-1009	TCS	10	₹1500.00	BUY	Executed	---	---			
ORD-1010	TCS	10	₹1500.00	SELL	Executed	---	---			
ORD-1011	WIPRO	10	₹1400.00	BUY	Executed	---	---			
ORD-1012	WIPRO	10	₹1400.00	SELL	Executed	---	---			
ORD-1013	RELIANCE	10	₹1400.00	BUY	Executed	---	---			
ORD-1014	INFY	10	₹1500.00	BUY	Executed	---	---			

Screencast Video

localhost:3000/Signup

YouTube Gmail YouTube Maps Translate Imported From Fir... sahil easy shortcu... Jobs/Intern Edutech Germany accomo... research_IU

New Chrome available

ZERODHA

SignupAboutProductPricingSupport

Create Account

Full Name

zuba

Email Address

you@example.com

Password

At least 6 characters

Confirm Password

Re-enter your password

Create Account

Already have an account? [Log In](#)

Note:

This is a simulation trading application. After signup, you'll be redirected

Summary

The Zerodha-like Trading Application is a full-stack web-based platform developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js) to simulate a real-world online trading environment. The application is designed to provide users with a responsive, secure, and user-friendly interface for analyzing financial markets, managing investments, and executing trades efficiently.

It follows a layered architecture where the React-based frontend handles user interaction and visualization, the Node.js and Express backend manages business logic and API communication, and MongoDB stores user data, portfolios, and transaction records. The platform includes key features such as real-time portfolio tracking, advanced charting for technical analysis, fast order execution, profit and loss monitoring, and detailed performance reports.

Additionally, it offers fund management capabilities, price alerts, and integrated tools for enhanced analytics. The user interface is divided into essential modules, including a homepage for onboarding and education, a dashboard for real-time portfolio insights with graphical representations, and a support portal for user assistance and issue resolution. The application also incorporates testing using Jest and React Testing Library to ensure reliability and proper functionality of UI components and trading operations.

Overall, this project demonstrates a realistic and scalable trading system, combining modern web technologies, clean architecture, and practical financial features, making it a strong representation of a professional trading platform suitable for both academic and portfolio purposes.

*Thank
you!*